Bathroom Monitoring with Fast-Chirp Modulation Millimeter-wave UWB Radar

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***Abstract*— Bathroom monitoring system has several problems such as privacy, multipath and humidity. High resolution radar will be effective to solve the problem in bathroom because of high range-resolution and Doppler-range resolution as well as low privacy and anti-humidity capability. In this paper, we propose a monitoring system based on hidden Markov model using fast-chirp modulation sensor in order to detect accidents caused by heat-shock. We perform experiment about three scenarios (normal, falling down and drowning.) and discus the estimation result by prior learning and incremental learning. As a result, the state detection rate is found to be more than 95 % on prior learning and approximately 90 % on incremental learning.**

***Index Terms* — millimeter-wave, radar, bathroom, fast-chirp modulation, direction of arrival, hidden Markov model.**

I. Introduction

The number of home accident of elderly is increasing with the arrival of super-aging society. Especially in the bathroom, an estimated 19,000 people have died due to heat shock or heat stroke. Please note that typical Japanese bathroom is equipped with shower and deep bathtub and the toilet is usually located in an entirely separate room. The bathtub is used for soaking and the bath water tends to be relatively hot. Some wireless sensors such as camera, IR sensor, and Doppler sensor are being considered as sensors, but they are difficult to use in bathrooms due to privacy infringement, humidity, and fluctuations in the water level in the bathtub. Therefore, there are a few papers about bathroom accident detection using UWB wireless sensor [1,2]. In [1], UWB wireless sensor using feature extraction was reported against waving from the surface of bathtub water. In [5], a bathroom accident detection system was presented using k-means. However, it has problem of classification result depends on the initial parameters. Over-fitting may be caused by individual and environment because the feature depends on individual. Therefore, basher is required to set the parameter to avoid over-fitting.

In this paper, we suggest a bathroom monitoring system using a 79GHz millimeter-wave UWB sensor with fast-chirp modulation (FCM) scheme and hidden Markov model (HMM). The FCM radar has multiple receive antenna, thereby estimate direction of arrival and doppler and can extract human movement on each range bin. Doppler signals due to human movement can be discriminated from the multipath of stationary objects. And HMM assumes a potential state to which a certain probability model is assigned and estimates the state transition from features. The behavior in the bathroom can be modeled by HMM. HMM is expected the stable classification result because the learning is based on the probability and a model. We discuss the algorithm for monitoring system and performed experiment. Please note that we discuss the case where the sensor installs on the wall along long side of bathtub.

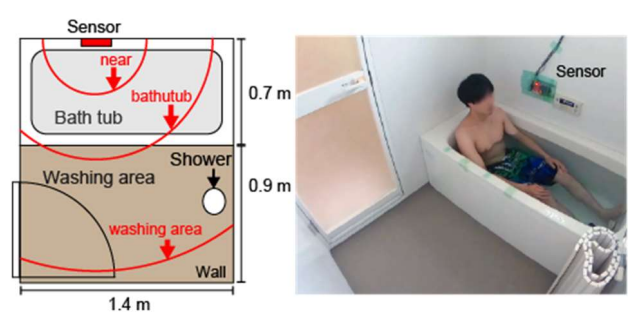
II. Feature Extraction

Millimeter-wave UWB radar offers high range resolution and anti-multipath. It can also be miniaturized. In this paper, the 79GHz millimeter-wave UWB radar is used for bathroom monitoring system. This sensor adopts FCM scheme. It transmits radio wave with frequency modulated and can acquire the relative velocity as well as the distance and angle from the sensor. Specifically, it is possible to extract the reflection intensity distribution in a three-dimensional space with distance, angle, and velocity as axes. Installing the sensor system on the wall, the longitudinal direction of the bathroom, range area can be defined for each distance shown in Fig.1. These are defined as neighborhood area (near area), bathtub area, washing area, and wall area. Their range calculate from range of wall R_{wall} as follows

$$R_{\mathrm{near}} = \frac{R_{\mathrm{wall}}}{4}, \tag{1}$$

$$R_{bath} = R_{\mathrm{near}} + \frac{R_{\mathrm{wall}}}{4}, \tag{2}$$

$$R_{\mathrm{wash}} = R_{\mathrm{bath}} + \frac{R_{\mathrm{wall}}}{2} - m, \tag{3}$$



(a) Bathroom model (b) State of experiment

Fig. 1. Range area definition and bathroom model.

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where m is range adjustment parameter in order to exclude wall signal in washing area. We calculate four features using the range: area standard deviation, area doppler feature, area direction feature and human motion feature.

A. Area standard deviation feature

Area standard deviation feature S represents small movement such as basher's motion. The standard deviation is calculated in each range bin of range spectrum as follows

$$\sigma_x(t,r) = \sqrt{\left(\frac{1}{N}\sum_{n=0}^{N}(x(t-n,r)-\bar{x})^2\right)}, \quad (4)$$

where x is range spectrum and r is index of range bin. We extract the feature in three area: near, bathtub, and washing. the features are calculated as follows

$$S_{\text{near}}(t) = \sum_{r=0}^{R_{\text{near}}-1}\sigma_x(t,r), \quad (5)$$

$$S_{\text{bath}}(t) = \sum_{r=R_{\text{near}}}^{R_{\text{bath}}-1}\sigma_x(t,r), \quad (6)$$

$$S_{\text{wash}}(t) = \sum_{r=R_{\text{bath}}}^{R_{\text{wash}}-1}\sigma_P(t,r). \quad (7)$$

B. Area doppler feature

Area doppler feature V represents human movement. It is calculated by range spectrum adapted moving target indication (MTI) in velocity dimension. MTI extracts human motion from range spectrum. We extract the feature in three area: bathtub, washing and wall. the features are calculated as follow

$$V_{\text{bath}}(t) = \sum_{r=0}^{R_{\text{near}}-1}x_{\text{mti}}(t,r) \quad (8)$$

$$V_{\text{wash}}(t) = \sum_{r=R_{\text{bath}}}^{R_{\text{wash}}-1}x_{\text{mti}}(t,r) \quad (9)$$

$$V_{\text{wall}}(t) = \sum_{r=R_{\text{wash}}}^{R_{\text{wall}}+m-1}x_{\text{mti}}(t,r) \quad (10)$$

C. Area direction feature

Area direction feature D represents direction of human movement. It is calculated by range spectrum adapted direction of arrival estimation (DOA) and MTI. We adapt DOA two angle and defined it $x_{\text{right}}(t,r)$ and $x_{\text{left}}(t,r)$. The area direction is calculated as follow

$$D_{\text{wash}}(t) = \max[x_{\text{right}}(t,r)] - \max[x_{\text{left}}(t,r)] \quad (11)$$

The direction is decided by value of $D_{\text{wash}}(\text{t})$. If the absolute value of $D_{\text{wash}}(\text{t})$ is very small, the direction is determined to be central.

D. Human motion feature

Human motion feature H extracts human motion in bathroom. it is calculated as follow

$$H(t) = \frac{S_{\text{wash}}(\text{t}) + aS_{\text{bath}}(\text{t})}{2} \quad (12)$$

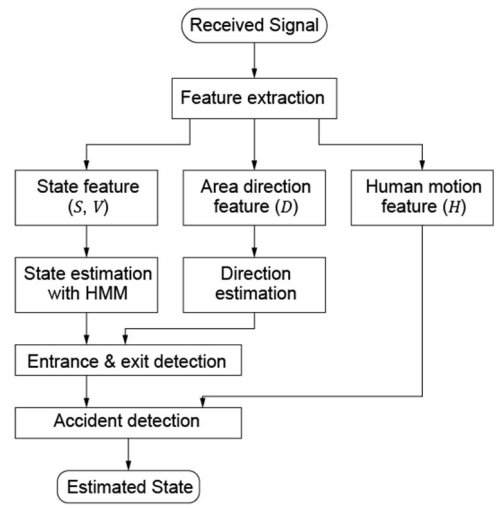

Fig. 2. A flowchart of bathroom monitoring system in order to various accident detection.

TABLE I
LIST OF ESTIMATED STATE AND GROUPS

State estimation result	Group
Empty	Empty
Entry and exit	Washing area
Washing area	
Movement in washing area	
Movement of bathing	
Movement during bathing	Bathtub
Bathing	

TABLE II
SCENARIOS DETAIL

Scenario A	Scenario B	Scenario C
Enter to the bathroom		
Showering in the washing area		
Soaking in the bathtub		Falling in the washing area
Getting out of the bathtub	Drowning in the bathtub	
Exit from the bathroom		

where a is a coefficient of standardization between S_{wash} and S_{bath}.

III. DETECTION ALGORITHM

Figure 2 shows flowchart of detection algorithm. State feature is composed of S and V. In state estimation, HMM estimates seven state as shown in Table I. In entry and exit detection, the sensor detects direction of an entrance. The direction of entry and exit must be the same. In typical Japanese bathroom, bashers do not enter from the center, thus detector does not determine that it is an entrance if the basher is in the center of bathroom. Accident detection

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determines some accident from the appearance of manned or unmanned. It is determined by thresholding H. The bather is determined to be in risk if the state that the result of HMM determines manned, but thresholding determines unmanned continues for a few second.

IV. Experiment and Result

We performed experimental measurement in a bathroom. The measurement environment is shown in Fig. 1. The parameters were a center frequency of 79 GHz, a bandwidth of 3.3 GHz, a transmission power of 0 dBm, and a sampling rate of 5.0 fps. Please note that the detection interval is 0.5 fps. We considered three scenarios shown in Table II. Scenario A is normal bathing without accident. So, no false alarm is required to occur. Scenario B and Scenario C are drowning in bathtub and falling down respectively. Scenario A was performed eight times. Scenario B and scenario C is performed four times its own. Six training data of scenario A was used to train.

Three subjects (male in their 20s), A, B and C, were measured according to the scenarios. We performed simple learning and incremental learning. In simple learning, training is performed using training data from the same subject. In incremental learning, when different subjects are measured, data from new subjects is added as new training data, and the existing model is retrained. The incremental learning is applied to improve estimation accuracy, and compared to existing model. The results are shown in Fig. 3 and Table III. The estimation accuracy rate (ER) is the estimation accuracy in which the four states shown in Table I. The danger detection rate (DR) is the detection accuracy of dangerous state in the state estimation. The false alarm rate (FR) is the rate at which it is judged to be dangerous in safety state. Average of ER and DR in estimation with same subjects are 95 % and 98 %, respectively. DR degradation was caused by delay of detection at the drowning. Also, FR is very small. However, when estimation performs for the different subject without incremental learning, ER and DR degrade. It is caused by individual of human motion feature. After incremental learning, SR is improved more than before the learning. It shows that the proposal can avoid the over-fitting by individual.

V. Conclusion

In this paper, we discussed bathroom monitoring system using 79 GHz UWB radar. We suggested an accident detection algorithm based on HMM and discussed about detection accuracy. As a result, the state detection rate was more than 95 % on prior learning and approximately 90 % on incremental learning.

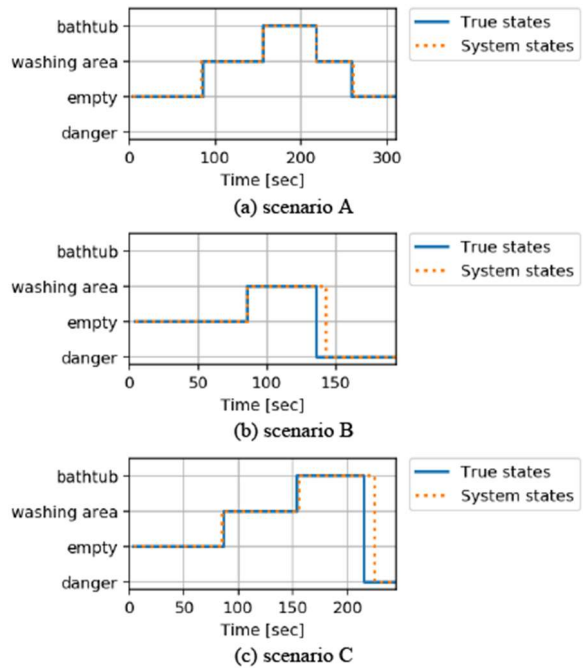

Fig. 3. An example of estimation result of a subject.

TABLE III
ESTIMATION ACCURACY

(a) estimation with same subject

subject	ER (%)	DR (%)	FR (%)
A	95.2	93.1	0.0
B	95.3	99.8	0.0
C	94.2	100.0	2.6

(b) estimation before incremental learning

model	subject	ER (%)	DR (%)	FR (%)
A	B	91.3	86.5	4.7
	C	86.8	68.6	1.1
B	A	94.8	100.0	0.4
	C	81.3	98.6	5.5
C	A	95.6	99.9	0.3
	B	90.9	100.0	6.6

(c) estimation after incremental learning

model	subject	ER (%)	DR (%)	FR (%)
A	B	95.0	89.8	0.1
	C	95.3	92.8	0.0
B	A	95.7	99.7	0.0
	C	89.7	97.7	0.0
C	A	95.7	99.5	0.0
	B	94.3	100.0	1.9

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